Prof. Olushina Olawale Awe

(Council Member of the International Statistical Institute (ISI))
Email: olushina-awe@ph-ludwigsburg.de, olawaleawe@gmail.com
Mobile: +4915203693944, +5519989659889, +2348034558906
LinkedIn: linkedin.com/in/O.Olawale Awe | Location: Ludwisburg, Germany |
Homepage: www.olawaleawe.org | Scopus h-index- 12:
https://www.scopus.com/authid/detail.uri?authorId=57194525020

PROFILE

Professor Olushina Olawale Awe is an internationally recognized statistician and data scientist with about 20 years of university-level teaching, research, and graduate supervision experience in machine learning, modelling, deep learning, and data science. He has authored 100+ peer-reviewed publications and edited major scholarly volumes published by Springer and Taylor & Francis. His research expertise spans functional and time-series data analysis, Bayesian and computational statistics, statistical learning, machine learning, health data science, and interpretable artificial intelligence, with applications in public health, economics, climate, and policy analytics. He currently serves as Vice President of Global Statistical Engagements with the LISA 2020 Global Network, headquartered at the University of Colorado Boulder, USA where he has led the establishment of international statistical consulting laboratories and large-scale research capacity-building initiatives that cut across 15 countries. He is also vice president of the International Association for Statistical Education and a council member of the International Statistical Institute, reflecting sustained scholarly leadership and international recognition. His research agenda emphasizes rigorous statistical foundations for machine learning, interpretable and responsible AI, and scalable data-driven methods, with a strong commitment to graduate training, interdisciplinary collaboration, and high-impact research.

CAREER EXPERIENCE

1. AvH Foundation Professor (Statistical and Data Science Literacy) -
   PH Ludwigsburg University, Germany (2025–Present).
   (Alexander von Humboldt Experienced Fellow).
2. Research Professor— Machine Learning Team Lead (09/2024 - Present)
   (Statistical Learning Laboratory, Federal University of Bahia (UFBA), Brazil)
3. Distinguished Professor of Data Science-
   Miva Open University, Abuja, Nigeria (2024- Present).
4. Associate Professor, Global Humanistic University, Curacao. (2021-2024).
5. Senior Lecturer (Assistant Professor)(09/2017 - 08/2021)
   Department of Mathematical Sciences, Anchor University Lagos, Nigeria.
6. Coordinator, Anchor University's Statistical Consulting Lab (09/2017 - 09/2021).
7. Lecturer (AL-L1), Department of Mathematics, Obafemi Awolowo University, Nigeria
   (Served as the Coordinator of the Laboratory for Interdisciplinary Statistical Consulting and Collaboration (LISAC)) (08/2010 - 09/2017)
8. Visiting J1 Scholar, Department of Statistics, Virginia Tech, USA
   (09/2013 - 09/2014)
9. Ag. Head, Accounts and Admin (03/2007 - 08/2010)
   WEMA Bank Plc, Ile-Ife, Nigeria

10. Mathematics Teacher (NYSC Scheme) (07/2004 - 08/2005)
    Federal Technical College, Jalingo, Nigeria

Professional Impact

Vice President, Global Statistical Engagements 2022–Present
LISA 2020 Global Network (USA)

- Coordinates and mentors 40 statistical and data science laboratories in 12 countries.
- Provides strategic and operational training and leadership to graduate students (MSc and PhD), faculty, and professional staff in advancing LISA's mission of interdisciplinary statistical collaboration, research infrastructure development, and statistical literacy outreach.
- Trained **225 statisticians** to serve as effective interdisciplinary statistical collaborators across diverse research domains.
- Personally collaborated on **250+ interdisciplinary research projects** spanning health, education, social sciences, and applied data science.
- Oversaw statistical support for **9,429 researchers**, including collaboration on **2,719 research projects**.
- Delivered **2,962 walk-in statistical consulting sessions** supporting researchers at various stages of the research lifecycle.
- Coordinated and delivered **196 short courses**, training **3,748 participants** in applied statistical methods, data analysis, and research design.

EDUCATION

1. University of Ibadan, Nigeria (2012 - 2016)
   PhD (Statistics)(Evaluated as a USA Equivalent Regional PhD by ECE)
2. University of Sao Paulo (USP), Brazil (2023-2025)
   MBA Data Science and Analytics.
3. Obafemi Awolowo University, Nigeria (2008 - 2010)
   Master of Business Administration (Finance)
4. University of Ibadan, Nigeria (2005 - 2007)
   M.Sc. (Statistics)
5. University of Ilorin, Nigeria (2000 - 2003)
   B.Sc. (Statistics)**(Second Class, Upper Divison)**
6. Federal Polytechnic Offa, Nigeria (1997-1999)
   National Diploma in Mathematics and Statistics (Upper Credit)

COMPLEMENTARY TRAINING

1. FAPESP International Research Fellowship (09/2021- 08/2023)
   (Institute of Mathematics and Statistical Computing), UNICAMP, Brazil

2. CAPES Postdoctoral Research Fellowship 03/2020 - 03/2021
   Institute of Mathematics and Statistics, Federal University of Bahia (UFBA),

Salvador, Brazil.

.

3. Virginia Tech, USA (09/2013 - 09/2014)
   Certificate of Excellence in Interdisciplinary Statistics and Data Science

Courses Taught

Courses I have taught include:

(a) **Data Literacy for Critical Thinking** Data Literacy for Critical Thinking is a transformative course designed to help learners interpret, question, and use data responsibly in everyday decision-making, education, and civic life. It introduces key concepts in data collection, visualization, statistical reasoning, and ethical interpretation, emphasizing how data can both inform and mislead. Through real-world examples, interactive exercises, and reflective analysis, participants develop the skills to identify biases, evaluate evidence, and make sound judgments based on data. The course empowers students to move beyond numbers—to think critically about the stories data tell, the power structures they reflect, and their implications for informed citizenship and social responsibility. I co-teach this course with colleagues at PH Ludwigsburg University of Education, Germany.

(b) **Artificial Intelligence in Education** Artificial Intelligence in Education is an interdisciplinary course that examines how AI technologies are transforming teaching, learning, and educational management. It introduces participants to the principles and applications of AI, such as adaptive learning systems, machine learning for critical thinking, intelligent tutoring, learning analytics, and generative AI, while addressing critical issues of ethics, bias, and equity in educational contexts. Through conceptual exploration and practical engagement with real-world tools, the course equips educators, researchers, and policymakers to critically evaluate, design, and implement AI-enhanced learning environments that foster personalization, inclusivity, and data-informed decision-making in modern education. I co-teach this course at PH Ludwigsburg University of Education, Germany.

(c) **Deep Learning Methods and Algorithms** This course provides a comprehensive and intuitive introduction to deep learning, equipping learners with both the theoretical foundations and practical skills needed to build modern neural network models. Using a structured and beginner-friendly approach, the course explores how deep neural networks learn from data, the mathematics behind key learning algorithms, and the architectural innovations that power today's AI systems. I developed and teach this course at the African Institute for Mathematical Sciences (AIMS).

(d) **Introduction to Statistics:** This introductory course provides students with a foundational understanding of statistical concepts. Topics taught include descriptive statistics, probability, and basic inferential statistics. The course serves as a primer for more advanced statistical coursework. (Taught undergraduate students at Anchor University Lagos and Obafemi Awolowo University, Nigeria).

(e) **Quantitative Methods** Quantitative Methods is a graduate-level course that equips students with the analytical tools and statistical frameworks necessary for evidence-based research and decision-making in complex, data-rich environments. The course introduces students to both the theoretical foundations and practical applications of quantitative analysis, with an emphasis on problem formulation, data interpretation, and the critical evaluation of research findings.

(f) **Stochastic Processes:** Stochastic Processes introduces students to the study of random processes evolving over time. Topics may include Markov chains, Poisson processes, and Brownian motion. The course explores the probabilistic modeling of dynamic systems. (I taught this course to undergraduate students at Obafemi Awolowo University, Nigeria).

(g) **Experimental Design:** Experimental Design focuses on the principles and methodologies of designing experiments to obtain meaningful and reliable results. Students learn about factors, randomization, and statistical techniques for planning and analyzing experiments. (Taught to senior undergraduate students at Obafemi Awolowo University, Nigeria).

(h) **Elementary Mathematics:** Elementary Mathematics covers fundamental mathematical concepts at a basic level. Topics may include arithmetic, algebra, geometry, and basic mathematical problem-solving skills. The course is designed to strengthen students' mathematical foundation. (Taught to undergraduate students at Anchor University Lagos and Obafemi Awolowo University, Nigeria).

(i) **Supervised Machine Learning:** This data science course explores the principles and applications of supervised machine learning algorithms. Students learn to train models on labeled datasets for tasks such as classification and regression. Common algorithms like decision trees, support vector machines, and neural networks were covered. It involves the use of R and Python for practical implementations. (I taught this course to MSc. students at the African Institute for Mathematical Sciences (AIMS), Cameroon. **Note: I developed this course from scratch.**)

(j) **Unsupervised Machine Learning:** Unsupervised Machine Learning introduces students to algorithms that do not require labeled training data. Clustering, dimensionality reduction, and association rule learning are among the topics covered. The course emphasizes pattern discovery in unlabeled datasets. (I taught this course to MSc. students at the African Institute for Mathematical Sciences (AIMS), Cameroon. **Note: I developed this course from scratch**.)

(k) **Statistical Decision Theory:** Statistical Decision Theory focuses on the principles of making decisions in the presence of uncertainty. Students learn about decision criteria, risk assessment, and the optimization of decision-making processes. The course combines statistical reasoning with decision analysis. (I taught this course to undergraduate students at Obafemi Awolowo University, Nigeria).

(l) **Applied Statistics and Statistical Computing:** This course explores the practical applications of statistical methods in various fields. Students learn to analyze and interpret data using statistical techniques, and the course covers topics such as regression analysis, data visualization, and statistical computing using tools like R or Python. (I taught this course to undergraduate students at Anchor University Lagos and Obafemi Awolowo University, Nigeria).

(m) **Probability Theory:** This course introduces students to the fundamental concepts of probability theory. Topics include probability distributions, conditional probability, and the laws of large numbers. The course aims to develop a solid understanding of uncertainty and randomness in various contexts. (Taught to undergraduate students at Anchor University Lagos and Obafemi Awolowo University, Nigeria).

(n) **Biostatistics and Statistical Inference:** Statistical Inference focuses on making inferences about populations based on sample data. Students learn hypothesis testing, confidence intervals, and the principles of estimation, sample size estimation, etc.

The course emphasizes the interpretation of statistical results in real-world scenarios. (I taught this course to undergraduate students at Anchor University Lagos and graduate level at Obafemi Awolowo University, Nigeria).

(o) **Advanced Mathematics:** This course delves into advanced mathematical concepts, potentially including topics from abstract algebra, advanced calculus, and linear algebra. It builds on foundational mathematical knowledge, introducing students to more abstract and complex mathematical structures. (Taught undergraduate students at Anchor University Lagos and Obafemi Awolowo University, Nigeria).

(p) **Data Science Fundamentals:** This course encompasses a range of topics to equip students with the knowledge and skills necessary for success in the field. The curriculum begins with foundational concepts in mathematics and statistics, covering areas such as probability, inferential statistics, and algebra. Students are then introduced to programming languages such as Python or R, enabling them to manipulate and analyze data. The course progresses to explore machine learning, covering both supervised and unsupervised algorithms for tasks like classification, regression, clustering, and dimensionality reduction providing a holistic and up-to-date approach to the dynamic field of data science. Throughout the course, hands-on projects and real-world applications are integrated to enhance practical skills and foster a deeper understanding of the complexities involved in extracting meaningful insights from diverse datasets. (Taught to undergraduate students at Obafemi Awolowo University, Nigeria).

(q) **Statistics in Economics (Econometrics):** This course integrates statistical methods into economic analysis. Students learn econometric techniques for analyzing economic data, estimating models, and testing hypotheses. The focus is on applying statistical tools to empirical economic problems. (Taught to undergraduate students from various social science departments at Anchor University Lagos.)

(r) **Regression Methods:** Regression Methods covers various regression models used in statistical analysis. Students learn to model relationships between variables, interpret coefficients, and make predictions. The course includes linear regression, logistic regression, and other regression techniques. (Taught to undergraduate students at Anchor University Lagos and Obafemi Awolowo University in Nigeria. I also taught it as a short course at Virginia Tech, USA in 2013).

(s) **Calculus:** This foundational course covers the principles of calculus, including differential and integral calculus. Topics include limits, derivatives, integrals, and applications in various fields. The course aims to develop students' understanding of the fundamental concepts of calculus and their applications. (Taught to undergraduate students at Anchor University Lagos and Obafemi Awolowo University, Nigeria.)

Academic Services

I rendered the following academic services at Anchor University Lagos.

- I coordinated AULISSDA Statistical Consulting Center for 4 years.
- Senior Lecturer in Statistics, Department of Mathematical Sciences: Courses taught include Applied Statistics and Statistical Computing, Probability, Statistical Inference, Advanced Mathematics, Mathematical Methods, Partial Differential Equations, Statistics in Economics (Econometrics), Calculus.
- I attracted Professor Eric Vance from the University of Colorado Boulder, Boulder USA to Anchor University Lagos to establish and inaugurate Anchor University Lab-

oratory for Interdisciplinary Statistical Science and Data Analysis (AULISSDA) on September 4, 2017.

- I generated internal revenues through AULISSDA workshops and seminars.
- Conducted research in Machine Learning, Biometrics, Environmental Science, Econometrics, and Time Series Analysis.
- I served as Adjunct Senior Lecturer to Departments of Economics, Business Administration and, Biological Sciences and Accounting.
- I attracted Four (5) erudite visiting scholars from the USA, Kenya, and Spain to present capacity-building seminars and workshops at Anchor University Lagos.(Prof. Eric Vance (USA), Monica Johnston (USA), Professor Afees Salisu (GHU, Curacao), Professor Robert Mudida (Kenya), Professor Luis Gil-Alana (Spain)).
- Represented AUL and presented some results of my research at the Centre for the Studies of African Economies 2018 Conference, Oxford University, United Kingdom.
- I Facilitated the signing of a Memorandum of Understanding on joint research and exchange of staff and students between Anchor University and the University of Colorado Boulder, USA.
- Organized the First International Workshop on Time Series Analysis in Nigeria (Sponsored by World Bank and ISBIS) at AUL.
- Revised Statistics syllabus to meet accreditation standards.
- Served as Panel Member, Mock Accreditation Exercise 2019- Anchor University, Lagos.
- Chair, Committee on Monthly Faculty of Science Colloquium.
- Served as Chair of the University Research and Development Committee.
- Member of the Senate Committee on Academic Planning and Quality Assurance.
- Appointed as the University SIWES Coordinator.
- Participated as a staff in the accreditation of Eleven (11) courses at the university.
- Member, Editorial Committee of Anchor University Journal of Science and Technology (AUJST).
- Projected the good image of the university to the international academic community.
- Chair, Data Science and Statistics Curriculum Committee.

RESEARCH ACTIVITIES/AREAS

- Statistical Modeling and Data Science
- Computational Statistics and Machine Learning
- Explainable AI methods with applications to health and environmental data.
- Time Series Econometrics
- Statistics Education and Interdisciplinary Research
- Functional Data Analysis with Machine Learning Integration

- Biostatistics

SCIENTIFIC PUBLICATIONS

Books and Edited Volumes

(a) Awe, O.O. (2026) Foundations of Civic Data Science—Forthcoming in Springer Nature, Switzerland.

(b) Awe, O. O., & Vance, E. A. (Eds.). (2024). Sustainable Statistical and Data Science Methods and Practices: Reports from LISA 2020 Global Network, Ghana, 2022. Springer International Publishing AG.

(c) Awe, O.O., and Vance, E.A. (2024). Practical Statistical Learning and Data Science Methods (Springer Nature, Switzerland).

(d) Awe, O.O., Love, K., and Vance, E. A. (2022). Promoting Statistical Practice and Collaboration in Developing Countries (Taylor and Francis).

(e) Awe, O. O. (2012): Econometric time series analysis: A quantitative focus on a developing economy. Lambert Academic Publishers, Germany. (Monograph).

Book Chapters

i. Awe, O. O., & Adepoju, J. M. (2024). Optimizing Machine Learning Models for Disease Diagnosis Using Bayesian Hyperparameter Optimization (BHO). In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 373-390). Cham: Springer Nature Switzerland.

ii. Panda, P., Vatsa, R., Chime, D., & Awe, O. O. (2024). Understanding Sentiment Analysis: Insights from Student Reviews of Universities and IMDb Movie Reviews. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 341-372). Cham: Springer Nature Switzerland.

iii. Diallo, R., Edalo, C., & Awe, O. O. (2024). Machine Learning Evaluation of Imbalanced Health Data: A Comparative Analysis of Balanced Accuracy, MCC, and F1 Score. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 283-312). Cham: Springer Nature Switzerland.

iv. Fandio, A., & Awe, O. O. (2024). Comparative Study of Supervised Machine Learning Algorithms for Predicting Oversampled Imbalanced Medical Data. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 667-696). Cham: Springer Nature Switzerland.

v. Edalo, C., Diallo, R., & Awe, O. O. (2024). Machine Learning Prediction of Multiclass Credit Score Classification. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 413-433). Cham: Springer Nature Switzerland.

vi. Agunloye, O. K., & Awe, O. O. (2024). Comparative Analysis of Lag Selection Effects on Unit Root Test Performance. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 195-204). Cham: Springer Nature Switzerland.

vii. Sinha, A. K., & Awe, O. O. (2024). CR Rao: A Legacy of Pioneering Contributions in Statistics and Data Science. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 741-752). Cham: Springer Nature Switzerland.

viii. Adepoju, J. M., Adetunji, E. T., & Awe, O. O. (2024). Predictive Modeling for Disease Diagnosis Using Calibrated Machine Learning: A Comparative Analysis of Spline, Beta, and Platt Calibration Scaling. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 435-462). Cham: Springer Nature Switzerland.

ix. Njoku, A. O., Mpinda, B. N., & Awe, O. O. (2024). Enhancing Predictive Performance Through Optimized Ensemble Stacking for Imbalanced Classification Problems. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 575-595). Cham: Springer Nature Switzerland.

x. Awe, O. O., Adepoju, J. M., Boniface, E., & Awe, O. D. (2024). Comparative Analysis of Random Forest and Neural Networks for Anemia Prediction in Female Adolescents: A LIME-Based Explainability Approach. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 555-573). Cham: Springer Nature Switzerland.

xi. Mwangi, P., Kotva, S., & Awe, O. O. (2024). Explainable AI Models for Improved Disease Prediction. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 73-109). Cham: Springer Nature Switzerland.

xii. Awe, O. O., & Dias, R. (2024). A Comparative Forecasting Study on Functional Temperature Data Across 32 American Countries. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 237-252). Cham: Springer Nature Switzerland.

xiii. Tsabedze, M., Mpinda, B. N., & Awe, O. O. (2024). Novel Sampling Methods for Imbalanced Data Prediction: Simulations and Applications to Medical Data. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 525-553). Cham: Springer Nature Switzerland.

xiv. Awe, O. O., Salako, J., Rodrigues, P. C., Dukhi, N., & Dias, R. (2024). A Comparative Exploration of SHAP and LIME for Enhancing the Interpretability of Machine Learning Models in Obesity Prediction. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 253-281). Cham: Springer Nature Switzerland.

xv. April, B. K., Oluoch, L., & Awe, O. O. (2024). Novel Applications of Bayesian Additive Regression Model for Predicting Diamond Prices: A Comparative Study of Tree-Based Ensemble Techniques. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 127-164). Cham: Springer Nature Switzerland.

xvi. Mamba, S., Djomou, F. R. K., & Awe, O. O. (2024). Explainability of Digital Wallets' Fraud Detection Algorithms: Comparative Analysis of SHAP and Permutation Feature Importance. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 627-665). Cham: Springer Nature Switzerland.

xvii. Awe, O. O., & Adedeji, B. A. (2024). Spline Calibration for Optimizing Supervised Machine Learning Algorithms in the Presence of Varying Imbalanced Data Ratios. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 205-235). Cham: Springer Nature Switzerland.

xviii. Oladeji, T. A., Adeyemo, B. T., & Awe, O. O. (2024). Predicting Precipitation Time Series Dynamics in West Africa Using Deep Learning Models. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 111-125). Cham: Springer Nature Switzerland.

xix. Awe, O. O., & Dias, R. (2024). Exponential Smoothing and Neural Networks for Climate Forecasting in Brazil: Insights and Change-Point Prediction. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 391-411). Cham: Springer Nature Switzerland.

xx. Maale, F. D., Okyere, G. A., & Awe, O. O. (2024). Effects of Imputation Techniques on Predictive Performance of Supervised Machine Learning Algorithms. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 29-48). Cham: Springer Nature Switzerland.

xxi. Adepoju, J. M., Opateye, G. O., & Awe, O. O. (2024). Decision Tree Pruning Strategies for Predicting Obesity. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 165-194). Cham: Springer Nature Switzerland.

xxii. Opateye, G. O., Johnson, C. A. G., Oyelahan, T. T., & Awe, O. O. (2024). Overweight Prediction Using Weighted Hard and Soft Voting Ensemble Machine Learning Classifiers. In Practical Statistical Learning and Data Science Methods: Case Studies from LISA 2020 Global Network, USA (pp. 49-71). Cham: Springer Nature Switzerland.

xxiii. Awe, O. O., Ojumu, J. B., Ayanwoye, G. A., Ojumoola, J. S., & Dias, R. (2024). Machine Learning Approaches for Handling Imbalances in Health Data Classification. In Sustainable Statistical and Data Science Methods and Practices: Reports from LISA 2020 Global Network, Ghana, 2022 (pp. 375-391). Cham: Springer Nature Switzerland.

xxiv. Adarabioyo, M. I., Adigun, K. A., Adejuwon, S. O., & Awe, O. O. (2024). Wealth Creation and Poverty Alleviation in a Nigerian State: A Recent Evidence-Based Survey. In Sustainable Statistical and Data Science Methods and Practices: Reports from LISA 2020 Global Network, Ghana, 2022 (pp. 91-110). Cham: Springer Nature Switzerland.

xxv. Awe, O. O., Love, K., & Vance, E. A. (2024). Teaching Data Science in Africa Via Online Team-Based Learning. In Sustainable Statistical and Data Science Methods and Practices: Reports from LISA 2020 Global Network, Ghana, 2022 (pp. 409-415). Cham: Springer Nature Switzerland.

xxvi. Ajayi, K., Ajayi, I. H., Awe, O. D., & Awe, O. O. (2024). Exploring Feature Selection and Supervised Classification Algorithms for Predicting Obesity Among Rural Women for Policy Decisions. In Sustainable Statistical and Data Science Methods and Practices: Reports from LISA 2020 Global Network, Ghana, 2022 (pp. 41-56). Cham: Springer Nature Switzerland

xxvii. Awe, O. O., Opateye, G. O., Johnson, C. A. G., Tayo, O. T., & Dias, R. (2024). Weighted Hard and Soft Voting Ensemble Machine Learning Classifiers: Application to Anaemia Diagnosis. In Sustainable Statistical and Data Science Methods and Practices: Reports from LISA 2020 Global Network, Ghana, 2022 (pp. 351-374). Cham: Springer Nature Switzerland.

xxviii. Sinha, A. K., Vatsa, R., Love, K., Awe, O. O., & Vance, E. A. (2024). The Intersection of Data and Statistics with Sustainable Development Goals. In Sustainable Statistical and Data Science Methods and Practices: Reports from LISA 2020 Global Network, Ghana, 2022 (pp. 393-408). Cham: Springer Nature Switzerland.

xxix. Awe, O. O., Oladeji, T. A., Adeyemo, B. T., Olowookere, O. P., Aminu, F. F., Abiona, O. S., ... & Ayeni, E. O. (2024). Exploring Practical Applications and Python Code Snippets for Supervised Machine Learning Classification Algorithms. In Sustainable Statistical and Data Science Methods and Practices: Reports from LISA 2020 Global Network, Ghana, 2022 (pp. 213-246). Cham: Springer Nature Switzerland.

xxx. Awopegba, O. E., & Awe, O. O. (2024). Re-examining Inflation and Its Drivers in Nigeria: A Machine Learning Approach. In Sustainable Statistical and Data Science Methods and Practices: Reports from LISA 2020 Global Network, Ghana, 2022 (pp. 57-78). Cham: Springer Nature Switzerland.

xxxi. Awe, O. O., Dias, R., Ajetunmobi, T. K., Ayeni, O. C., Oyanameh, O. E., & Agunloye, O. K. (2024). Time Series Forecasting of Seasonal Non-stationary Climate Data: A Comparative Study. In Sustainable Statistical and Data Science Methods and Practices: Reports from LISA 2020 Global Network, Ghana, 2022 (pp. 335-350). Cham: Springer Nature Switzerland.

xxxii. Oluchukwu Njoku, A., Nyunga Mpinda, B., & Awe Olawale O. (2023). Improving the Accuracy of Financial Bankruptcy Prediction Using Ensemble Learning Techniques. In Pan African Conference on Artificial Intelligence (pp. 3-29). Cham: Springer Nature Switzerland.

xxxiii. Awe, O. O., Fagbamigbe, A. F., Evans, E., Olorukooba, A. A., Awe, O. D., Adarabioyo, M. I., ... & Babalola, B. T. (2022). Building Biostatistics Capacity in Developing Countries. In Promoting Statistical Practice and Collaboration in Developing Countries (pp. 495-514). Chapman and Hall/CRC.

xxxiv. Vance, E. A., Love, K., Awe, O. O., & Pruitt, T. R. (2022). LISA 2020: Promoting Statistical Practice and Collaboration in Developing Countries. In Promoting Statistical Practice and Collaboration in Developing Countries (pp. 27-46). Chapman and Hall/CRC.

xxxv. Babalola, B. T., Awe, O. O., & Adarabioyo, M. I. (2022). Challenges of Statistics Education That Leads to High Dropout Rate among Undergraduate Statistics Students in Developing Countries. In Promoting Statistical Practice and Collaboration in Developing Countries (pp. 75–83). Chapman and Hall/CRC.

xxxvi. Awe, O. O., & Vance, E. A. (2022). The ABC of Successful Statistical Collaborations: Adapting the ASCCR Frame in Developing Countries. In Promoting Statistical Practice and Collaboration in Developing Countries (pp. 149-156). Chapman and Hall/CRC

xxxvii. Awoleye, O. M., Abegunde, A. A., & Awe, O. O. (2022). Retrieval of Unstructured Datasets and R Implementation of Text Analytics in the Climate Change Domain. In Promoting Statistical Practice and Collaboration in Developing Countries (pp. 285-299). Chapman and Hall/CRC.

xxxviii. Awe, O. O., Jegede, P. O., & Cochran, J. (2022). A Comprehensive Tutorial on Factor Analysis with R: Empirical Insights from an Educational Perspective. Promoting Statistical Practice and Collaboration in Developing Countries, 265. Chapman and Hall/CRC.

xxxix. Love, K., Awe, O. O., Gunderman, D. J., Druckenmiller, M., & Vance, E. A. (2022). LISA 2020 Network Survey on Challenges and Opportunities for Statistical Practice and Collaboration in Developing Countries. Promoting Statistical Practice and Collaboration in Developing Countries, 47-59. Chapman and Hall/CRC.

xl. Gil-Alana, L. A. and Awe, O.O. (2018): Persistence, structural breaks and non-linearities in stock price time series: empirical evidence from Nigeria. Advances in Financial Planning and Forecasting Vol. 9(2018), pp.1-15.

Journal Article Publications

i. Awe, O. O., Engel, J., & Weber-Stein, F. (2026). A Curriculum Framework for Integrating Large Quantitative Models into Statistical and Civic Data Science Education- Forthcoming in Statistics Education Research Journal. (Scopus)

ii. Awe, O. O., Mwangi, P. N., Goudoungou, S. K., Esho, R. V., & Oyejide, O. S. (2025). Explainable AI for enhanced accuracy in malaria diagnosis using ensemble machine learning models. BMC Medical Informatics and Decision Making, 25(1), 162. (Springer)

iii. Awe, O. O., Olaniyan, O. A., Olatunde, A. E., SewPaul, R., & Dukhi, N. (2025). A comparative analysis of generalized additive models for obesity risk prediction. Healthcare Analytics, Vol. 8(2005) 100410. (Elsevier)

iv. Muriithi, D. K., Lumumba, V. W., Awe, O. O., & Muriithi, D. M. (2025). Explainable Artificial Intelligence Models for Predicting Malaria Risk in Kenya. European Journal of Artificial Intelligence and Machine Learning, 4(1), 1-8.

v. Atofarati, E. O., Sharifpur, M., Huan, Z., Awe, O. O., & Meyer, J. P. (2025). Experimental and machine learning study on the influence of nanoparticle size and pulsating flow on heat transfer performance in nanofluid-jet impingement cooling. Applied Thermal Engineering, 258, 124631. (Elsevier)

vi. Awe, O. O., Atofarati, E. O., Adeyinka, M. O., Musa, A. P., & Onasanya, E. O. (2024). Assessing the factors affecting building construction collapse casualty using machine learning techniques: a case of Lagos, Nigeria. International Journal of Construction Management, 24(3), 261-269. (Taylor and Francis)

vii. Awe, O.O., Adedeji, B.A., and Dias, R (2024). Improving class probability estimates in asymmetric health data classification: An experimental comparison of novel calibration methods. Brazilian Journal of Biometrics 43(2). (Scopus)

viii. Obasi, S.N.N., Pemberton, J., and Awe, O.O. (2025). A Comparative Study of Soil Classification Machine Learning Models for Construction Management. International Journal of Construction Management. 28(9). (Taylor and Francis)

ix. Wanjohi, J.W., Mpinda, B. N. and Awe, O.O. (2024). Comparing logistic regression and neural networks for predicting skewed credit score: a LIME-based explainability approach. Statistics in Transition new series- Journal of the Polish Statistical Association, 25(3):49-67. (Scopus)

x. Salako, J., Ojo, F., and Awe, O. O. (2024). Fish-NET: Advancing Aquaculture Management through AI-Enhanced Fish Monitoring and Tracking. AGRIS Online Papers in Economics and Informatics, 16(2), 121-131. (Scopus)

xi. Awe, O.O., Dukhi, N., Dias, R (2023). Shrinkage Discriminant Algorithms for Classifying High-Dimensional Heteroscedastic Data: Insights from a National Health Survey. Machine Learning with Applications 12 (2023), 100459 (Elsevier).

xii. Sewpaul, R, Awe, O. O., Dogbey, D. M., Sekgala, D., & Dukhi, N. (2023). Classification of Obesity among South African Female Adolescents: Comparative Analysis of Logistic Regression and Random Forest Algorithms. International Journal of Environmental Research and Public Health, 21(11), 9757. (Scopus)

xiii. Awe, O. O., Atofarati, E. O., Adeyinka, M. O., Musa, A. P., & Onasanya, E. O. (2023). Assessing the factors affecting building construction collapse casualty using machine learning techniques: a case of Lagos, Nigeria. International Journal of Construction Management, 24(3), 261–269. (Taylor & Francis)

xiv. Awe, O. O., Musa, A. P., & Sanusi, G. P. (2023). Revisiting economic diversification in Africa's largest resource-rich nation: Empirical insights from unsupervised machine learning. Resources Policy, 82, 103540. (Elsevier).

xv. Fadeyi, T. E., Oyedemi, O. T., Awe, O. O., & Ayeni, F. (2023). Antibiotic use in infants within the first year of life is associated with the appearance of antibiotic-resistant genes in their feces. PeerJ. (Taylor & Francis)

xvi. Gayawan, E., Somo-Aina, O., & Awe, O.O (2022). Spatio-temporal dynamics of child mortality and relationship with a macroeconomic indicator in Africa. Applied spatial analysis and policy, 15(1), 143-159. (Springer)

xvii. Obidi, O. F., Awe, O. O., Vikesland, P., Leng, W., Chan, M., Oyagha, B., ... & Soyinka, O. (2022). Vibrational spectroscopy and non-parametric analysis for biodegradation evaluation: The acrylic-emulsion paint scenario. Vibrational Spectroscopy, 123, 103427 (Elsevier).

xviii. Awe, O. O. and Dias, R. (2022). Comparative Analysis of ARIMA and Artificial Neural Network Techniques for Forecasting Non-Stationary Agricultural Output Time Series", AGRIS online Papers in Economics and Informatics, Vol. 14, No. 4, pp. 3-9 (Scopus).

xix. Awe, O. O., Dogbey, D. M., Sewpaul, R., Sekgala, D., and Dukhi, N. (2021). Anaemia in children and adolescents: A Bibliometric analysis of BRICS countries (1990–2020). International Journal of Environmental Research and Public Health, 18(11), 5756. (Scopus)

xx. Awe, O.O., Mudida, R. and Gil-Alana, L. A. (2021): Comparative analysis of economic growth in Nigeria and Kenya: A fractional integration approach. International Journal of Finance and Economics (Wiley) 25(3). (Wiley).

xxi. Dukhi, N., Sewpaul, R., Sekgala, M. D., and Awe, O. O. (2021). Artificial intelligence approach for analyzing anaemia prevalence in children and adolescents in BRICS countries: a review. Current Research in Nutrition and Food Science Journal, 9(1), 01-10. (Scopus)

xxii. Awe, O. O., and Gil-Alana, L. A. (2021). Fractional integration analysis of precipitation dynamics: empirical insights from Nigeria. Tellus A: Dynamic Meteorology and Oceanography, 73(1), 1-9. (Taylor and Francis).

xxiii. Awe, O.O. Rodrigues, P.C., and Mahmoudvand, R. (2020): Non-negative time series reconstruction using singular spectrum analysis: A case study of precipitation dynamics in Nigeria. Fluctuation and Noise Letters (Elsevier-Scopus).

xxiv. Awe, O.O., Aromolaran, O. and Adepoju, A.A. (2020). Time-varying predictors of non-oil export volatility in Nigeria. African Journal of Business and Economic Research (AJBER), 16(1). Pp. 151-156. (Scopus)

xxv. Gayawan, E., Awe, O. O., Oseni, B. M., Uzochukwu, I. C., Adekunle, A., Samuel, G., ... and Adegboye, O. A. (2020). The spatiotemporal epidemic dynamics of COVID-19 outbreak in Africa. Epidemiology and Infection, 148, e212. (Cambridge University Press)

xxvi. Awe, O. O. and Adepoju, A. A. (2020): Bayesian estimation of time-varying parameters in the presence of discounted evolution variance. Journal of the Nigerian Mathematical Society 39(2): 183-209.

xxvii. Rodrigues, P. C., Awe, O. O., and Sousa, J. (2020): Modelling the behavior of currency exchange rates with Singular Spectrum Analysis and Artificial Neural Networks. Stats, 3(2), 137-157. (MDPI).

xxviii. Awe, O. O. and Adepoju, A. A. (2020): Change point detection in C02 emission-energy consumption nexus using a recursive Bayesian estimation approach. Statistics in Transition- Journal of the Polish Statistical Association, 21 (1):123-136. (Scopus)

xxix. Awe, O.O. and Gil-Alana, L. A. (2019): Time series analysis of economic growth rate series in Nigeria: structural breaks, non-linearity and reasons behind the recent recession. Applied Economics. 51 (50):5482-5489. (Taylor and Francis).

xxx. Awe, O.O, Akinlana, D.M., Yaya, O.S. and Aromolaran, O. (2018): Time series analysis of the behavior of import and export of agricultural and non-agricultural goods in West Africa: A case study of Nigeria. Agris On-line Journal of Economics and Informatics 10(2):15- 22.(Elsevier- Scopus).

xxxi. Awe, O. O. and Adepoju, A. A. (2018): Modified recursive Bayesian algorithm for estimating time-varying parameters in dynamic linear models. Statistics in Transition, Journal of the Polish Statistical Association, 19(2):58-82. (Scopus)

xxxii. Obidi, O.F., Awe, O.O., Igwo-Ezikpe, M.N. and Okekunjo, F. O. (2018): Production of Phosphatase by Microorganisms isolated from discolored painted walls in a typical tropical environment: A non-parametric analysis. Arab Journal of Basic and Applied Sciences. 25(3): 111-121 (Taylor and Francis).

xxxiii. Obidi, O. F., Awe, O. O., Okekunjo, F. O., and Igwo-Ezikpe, M. N. (2018). Studying the cellulolytic activity of microorganisms isolated from stained painted walls with reference to certain factors: a cross-sectional study. International Journal of Environmental Studies, 75(5), 740-749. (Taylor and Francis).

xxxiv. Onyeuwaoma, N. D., Chineke, T. C., Nwofor, O. K., Crandell, I., Awe, O. O., Olasumbo, A.,... and Joy, N. (2018): Characterization of Aerosol loading in urban and sub-urban locations: Impact on Atmospheric Extinction. Cogent Environmental Science, 4(2). (Taylor and Francis).

xxxv. Gil-Alana, L. A, Yaya, O. S., and Awe, O.O. (2017): Time series analysis of co-movements in the prices of Gold and Oil: A fractional co-integration approach. Resources Policy 53(117-124) (Elsevier).

xxxvi. Awe, O.O. (2012): On Pairwise Granger causality and econometric analysis of selected economic indicators. Interstat Online Journal (Virginia Tech, USA). Vol. 17: 1-17.

Selected Conference Proceedings

i. Awe, O. O., Engel, J., & Weber-Stein, F. (2026, February). Advancing critical data science and statistical literacy for democratic and civic education. In IASE Conference Proceedings Series, 2026.

ii. Weber-Stein, F., & Awe, O. O. (2026, February). Civic statistics in teacher education-Effects among future math and civic teachers. In IASE Conference Proceedings Series, 2026.

iii. Awe, O., Okeyinka, A., & Fatokun, J. O. (2020, March). An alternative algorithm for ARIMA model selection. In 2020 international conference in mathematics, computer engineering and computer science (ICMCECS) (pp. 1-4). IEEE.

iv. Adarabioyo, I., & Awe, O. O. (2020, March). Application of Bayesian tobit regression to global radiation. In 2020 International Conference in Mathematics, Computer Engineering and Computer Science (ICMCECS) (pp. 1-4). IEEE.

v. Awe, O., and Vance, E. (2018): Improving the teaching and learning of statistics in Nigerian universities through the LISA 2020 program. Proceedings of the Tenth International Conference on Teaching Statistics (ICOTS10, July 2018), Kyoto, Japan.

vi. Awe, O. O., Crandell, I., & Vance, E. A. (2015). Building statistics capacity in Nigeria through the LISA 2020 Program. In Proceedings of the International Statistical Institute's 60th World Statistics Congress. Rio de Janeiro, Brazil.

vii. Crandell I, Awe, O.O and Vance, E. A.(2015): LISA 2020: Developing Statistical Collaboration Capacity in Nigeria: Joint Statistical meetings, Seattle, USA

viii. Awe, O.O. and Vance, E.A. (2014): Statistics education, collaborative research, and LISA 2020: a view from Nigeria. Proceedings of the Ninth International Conference on Teaching Statistics (ICOTS9, July 2014), Flagstaff, Arizona, USA.

ix. Awe, O.O. and Vance E.A. (2014) : LISA 2020: A Vision Towards Research Development in Nigeria. Joint Statistical Meetings, Boston, MA. USA.

Other Articles

- Awe, O. D., Kipruto, H., Awe, O., & Chukwudum, Q. C. (2023). Trend Analysis of Maternal Mortality in Kenya: Post-Devolution Empirical Results. EA Health Research Journal, 7(2), 256-264.
- Obidi, O. F., Awe, O. O., Shogbade, A. O., & Akanji, A. F. (2022). Molecular Characterization of Successive Yeasts Strains and their Optimal Invertase Producing Conditions in Fresh Palm Wine (Raphiahookeri) Obtained from Lagos,

Nigeria. Journal of Applied Sciences and Environmental Management, 26(12), 2125-2134.

- Obidi, O. F., Awe, O. O., Igwo-Ezikpe, M. N., & Okekunjo, F. O. (2022). Empirical analysis of amylolytic and proteolytic activities of microbial isolates recovered from deteriorating painted wall surfaces in Lagos Nigeria. Bio-Research, 20(1), 1484-1496.
- Awe, O.O, Ayeni, O., Sanusi, G., and Oderinde, L. (2021). A comparative time series analysis of crude mortality rate in the BRICS countries. BRICS Journal of Economics, 2(2), 17-32.
- Awe, O. O., Aromolaran, O., and Adedayo, A. Time-Varying Model for Non-Oil Export Volatility in Nigeria. Sri Lankan Journal of Applied Statistics, 20, 2.
- Adarabioyo, I., Awe, O. O., Agunloye, O. K., and Gil-Alana, L. A. (2020). A novel approach for Stock Price Modeling based on fractional integration analysis. Solid State Technology, 63(6), 4467-4476.
- Badu, K., Thorn, J. P., Goonoo, N., Dukhi, N., Fagbamigbe, A. F., Kulohoma, B. W., Awe, O.O., ... and Gitaka, J. (2020). Africa's response to the COVID-19 pandemic: A review of the nature of the virus, impacts and implications for preparedness', AAS Open Research 3, 19.
- Aderemi, T.A., Awe, O.O, Adeyeye, T.A. and Pereowei, A.B. (2019): Panel cointegration and Granger causality approach to foreign direct investment and economic growth in South Asian Countries. Acta Universitatis Danubius. 15(7).pp.43-54.
- Olabiyi, K. A., and Awe, O. O. (2017). Children's Quality and Quantity in Nigeria: A Granger Causality Analysis. Humaniora, 8(3), 211-218.
- Adesunkanmi, S.O. and Awe, O.O. (2017): Capital base and micro business performance in Nigeria: A bootstrap regression approach. Canadian International Journal of Social Science and Education. 10(2017).
- Awe, O.O., Sanusi, K.A. and Adepoju, A. A. (2017): An application of Bayesian dynamic linear model to Okun's Law. Theoretical Mathematics and Applications 7(3): 27-43.
- Awe. O.O. Akinlana, D.M. and Adesunkanmi, S.O. (2016): Foreign trade-foreign exchange nexus in Nigeria: A vector error correction modeling approach. Binus University Business Review, 7(1):1-7.
- Awe, O. O. and Adepoju, A. A. (2015): Bayesian optimal filtering in dynamic linear models: An empirical study of economic time series data. British Journal of Mathematics and Computer Science. 7(6): 419-428.
- Awe, O.O., I. Crandell, A.A. Adepoju and S. Leman (2015): A time-varying parameter state-space model for analyzing money supply-economic growth nexus. Journal of Statistical and Econometric Methods. 4(1): 73-95
- Awe, O. O. (2015): Predictive dynamic linear models for external reserves-economic growth nexus: A case study. Journal of Applied Mathematics and Bioinformatics. 5(4): 81-98.
- Adelodun, O. A and Awe, O. O. (2013): Using LISREL program for empirical research. Transnational Journal of Science and Technology, 3(8): 1-14.

- Adarabioyo, M. I., Awe, O. O. and Lawal, F. K. (2012): Modelling socioeconomic factors affecting age at marriage among females in Kogi State, Nigeria. Mathematical Theory and Modeling, 2(2): 44-50.
- Oguntuase, D. M., Awe, O. O., & Ajayi, I. A. (2013). Empirical nexus between teaching/learning resources and academic performance in mathematics among pre-university students in Ile-Ife, South-West Nigeria. International Journal of Scientific and Research Publications, 3(3), 1-6.
- Awe, O. O., & Oguntuase, D. M. (2013). Acquisition and Utilization of Statistical Consulting and Collaboration Skills among University Students in Nigeria: A Recent Survey. International Journal of Computer and Electronics Research (India), 2(2).
- Adelodun, O. A., & Awe, O. O. (2013). Statistics education in Nigeria: A recent survey. Journal of Education and Practice, 4(11).
- Awe, O. O. (2012). Correlation or causality: new evidence from crosssectional statistical analysis of autocrash data in Nigeria. Transnatl J Sci Technol, 2(7), 48-63.
- Awe, O.O. (2012): How bad is multicollinearity: Evidence from multiple linear regression analysis. Int. Journal of Current Scientific Research, 2(2): 344-349.
- Awe, O. O. (2012): Fostering the practice and teaching of statistical consulting among young statisticians in Africa. Journal of Education and Practice, 3(3):54-59.
- Oni, T. O. and Awe, O. O. (2012): Empirical Nexus between Corruption and Economic Growth (GDP): A cross-country econometric analysis. International Journal of Scientific and Research Publications. 2(8): 8-14.
- Awe, O. O., Adarabioyo, M. I., Lawal, F. K., & Damola, A. (2012). Statistical Analysis of Some Socioeconomic Factors Affecting Age at Marriage among Males in Kogi State, Nigeria.
- Awe, O. O., & Adarabioyo, M. I. (2011). Multivariate regression techniques for analyzing auto-crash variables in Nigeria. Journal of Natural Sciences Research, 1(1), 19-34.

Media Publications

The following links contain some press coverages of my career activities:

- https://statsandstories.net/education1/lisa-collaborations - Stat + Stories
- https://guardian.ng/features/anchor-varsity-inaugurates-lab-centre-to-boost-research-data-analysis/
- https://theconversation.com/profiles/olushina-olawale-awe-1470282
- https://pmnewsnigeria.com/2019/05/14/anchor-universitys-lecturer-olawale-awe-gets-international-appointment/
- https://www.latestnigeriannews.com/news/7202604/anchor-university-lecturer-bags-international-appointment.html
- https://independent.ng/anchor-university-lecturer-gets-international-appointment/
- https://pmnewsnigeria.com/2017/12/10/anchor-univrsity-trains-researchers-latex/

- https://www.afidep.org/ela-training-workshop-another-cohort-of-research-to-policy-champions/
- https://www.afidep.org/multimedia/researchers-speak-on-their-plans-to-champion-evidence-use-by-governments/
- https://punchng.com/group-seeks-economic-improvement-through-data/

Participation in congresses, seminars, and similar events

Invited Talks

i. Teaching Interpretable AI and Critical Data Literacy in Civic Education- Paderborn University Colloquium on Artificial Intelligence and Data Science Education at School Level. (19 November 2025)

ii. Civic-Ready AI: Interpretable Machine Learning for Accountable Governance-Presented at AvH Network Meeting, Bayreuth, Germany (Nov 13, 2025)

iii. Data-Driven Leadership in the Fourth Data Revolution- Miva Open University Executive in Residence Live Session (Nov 8, 2025)

iv. Teaching Interpretable AI and Critical Data Literacy in Civic Education-Paderborn Colloquium on Artificial Intelligence and Data Science Education at School Level. (Nov 19, 2025)

v. Machine Learning for Critical Thinking in Educational Contexts- Presented during AI in Education Seminar- (Nov 19, 2025)

vi. Future Trends in Artificial Inteligence for Disease Modeling: Delivered at Center for Data Analytics and Modeling, Chuka University, Kenya - 09-08-2025.

vii. Explainable AIModels for Accurate Predictive Analysis-3rd International Conference on Statistical Theory, Methods and Applications (ICSTMA), 2025, Nsuka, Nigeria

viii. Wavelets Meet Machine Learning: A Multiresolution Perspective on Predictive Modeling. Inaugural Talk presented to all Six AIMS centers of the African Institute for Mathematical Sciences (AIMS)- 16.04. 2025.

ix. Introduction to Machine Learning in R. (Department of Applied Mathematics, UCBoulder, USA- October 9, 2024).

x. Computational Strategies for Handling Imbalanced Health Data in Machine Learning-Based Models, Training delivered at the AMMnet Annual Meeting, Kigali, Rwanda. (April 18-19 2024).

xi. Computational Strategies for Tackling Imbalanced Data in Machine Learning, Talk at International Association for Statistical Computing (IASC) (February 2024).

xii. Teaching Data Science in Africa via Online Team-Based Learning. Talk given at IASE Satellite Conference, Ottawa Canada (July 11, 2023).

xiii. Improving Probability Calibration in Machine Learning Models (Talk delivered at CLAPEM 2023, Sao Paulo Brazil).

xiv. Career Advice for Climate Scientists: Talk given at Climatematch Academy, USA (July 24, 2023).

xv. Supervised Discriminant Algorithms for High Dimensional Heteroscedastic Data (STODAD Seminar 2022, IMECC, UNICAMP).

xvi. Non-Parametric Trend Analysis and Autoregressive Neural Network Modelling of Clustered Climate Data (Latin American Conference on Statistical Computing-EBEB- LASC-2022).

xvii. Modified Shrinkage Discriminant Algorithms for High-Dimensional Heteroscedastic Data Classification (SINAPE, 2022-Gramado, Brazil).

xviii. Modified Shrinkage Heteroscedastic Discriminant Algorithms for Multi-Class High-Dimensional Survey Data- UFBA Seminar, UFBA, Salvador Brazil.

xix. Teaching Data Science via Virtual Team-Based Learning- LISA 2020 Sustainability Symposium, Kumasi Ghana- May 2022.

xx. Sensitivity Analysis of Shrinkage-Based Discriminant Algorithms for Classifying High-Dimensional Heteroscedastic Data: Insights from a National BMI Survey-RBRAS

xxi. Comparative Analysis of Machine Learning Algorithms for the Classification of Qutelet Index. 3rd Conference on Statistics and Data Science, UFBA, Brazil (28-30, Oct. 2021).

xxii. Time-varying Predictors of Export Volatility in Africa's Largest Economy. ICSDA 2019 Conference, Department of Economics, Covenant University Ota, Nigeria. (March '19).

xxiii. Bayesian Spatiotemporal analysis of child mortality in Africa. Talk delivered at AFIDEP/AAS Research to Policy Workshop, 2020, Nairobi, Kenya.

xxiv. Bayesian Estimation of Dynamic Linear Models with Time-Varying Parameters. Presented at the Fourth Faculty of Science and Science Education Colloquium, Anchor University Lagos. (21st May 2019).

xxv. Global Warming: Predicting C02 emission from Energy consumption in Africa. Talk delivered at COMA 2019, Nairobi Kenya.

xxvi. "LISA 2020: A Vision towards Research Development in Nigeria." Joint Statistical Meetings, Boston, MA, USA (August 2014).

xxvii. Statistics education, collaborative research, and LISA 2020: a view from Nigeria. (ICOTS9, July 2014), Flagstaff, Arizona, USA.

xxviii. A Time-Varying Parameter State-Space Model for the Nigerian Economy - Virginia Academy of Science Annual Conference, Richmond, Virginia. (16th May 2014).

xxix. Building Statistics Capacity in Nigeria through the LISA 2020 Program. 60th World Statistics Congress, Rio de Janeiro, Brazil. (July 2015).

xxx. How to Fit and Interpret your Regression Models. An invited talk at the Monthly One Hour with a Statistician Seminar at the Department of Statistics, University of Ibadan, Ibadan, Nigeria. (Tuesday, November 14, 2017).

Conferences Attended

- Joint Mathematical Meetings- Washington DC, USA (January 4-7, 2026)

- Alexander von Humboldt Network Meeting—Bayreuth, Germany (12-14 Nov, 2025)
- IASE Satellite Conference, University of Munster, Germany (1-2, October, 2025)
- 65th World Statistics Congress held at The Hague, Netherlands (5-9, October, 2025)
- 4th LISA 2020 Global Network Virtual Symposium (October 20-24, 2025)
- Institute for Disease Modeling Annual Symposium, Seattle, USA. October 1-2, 2024.
- R Medicine 2024 Virtual Conference, June 10-14, 2024.
- First International Day of Statistical Literacy (IDSL)-Virtual, May 21, 2024.
- AMMnet Kigali Annual Meeting, Marriot Hotel, Rwanda. April 18-19, 2024.
- IASE Satellite Conference- Ottawa, Canada (Virtual). July 11-13, 2023.
- R Medicine 2023 Virtual Conference, June 5-9, 2023.
- 16th CLAPEM- Sao Paulo, Brazil. July 10 - 14, 2023.
- 16th Brazilian Virtual Meeting of Bayesian Statistics and VI Latin American Conference on Statistical Computing-16-18 March 2022.
- 24th National Symposium on Probability and Statistics (SINAPE 2022), Gramado, Brazil. August 1-5, 2022.
- Short Course on Machine Learning Applications to Time Series, Majestic Palace Hotel, Florianopolis Santa Catarina, Brazil - 16-17 Nov. 2022.
- The 66th RBras Annual Meeting- (Bio) statistics and Biometrics in the age of Data Science- Florianopolis/Santa Catarina, Brazil-Nov. 16-18, 2022.
- 3rd Virtual Conference on Statistics and Data Science (CSDS)-Oct. 28-30, ‘21.
- 63rd Virtual World Statistics Congress July 22-26, 2021
- How to create and sustain a successful collaboration-virtual webinar organized by the African Academy of Sciences, Nairobi, Kenya (June 11, 2020) (Co-Facilitator). Virtual COVID-19 Innovation and Investment Forum 2020- June14-19, 2020.
- Biostatistics for Big Data Applications- University of Texas Medical Branch (UTMB) Virtual (September 2020) TX, USA.
- Capacity building training workshop in research communication/translation and policy engagement, Nairobi, Kenya (27-31 January 2020).
- Connecting Minds Africa Conference, United Nations Avenue, Nairobi, Kenya (25- 27 September 2019).
- 62nd ISI World Statistics Congress held at Palais Des Congress, Kuala Lumpur Convention Centre, Malaysia (18-23 August 2019).
- 2nd LISA 2020 International Symposium, Kuala Lumpur, Malaysia (August 15-18, 2019).
- Coastal Ocean Environmental Summer School in Ghana (COESSING 2019), Regional Maritime University, Ghana, (August 5-10, 2019).

- National Workshop on Article Writing and Research Publication held at Anchor University Lagos (28-29 March 2019) (Co-facilitator).
- 2019 International Conference on Sustainable Development in Africa (ICSDA 2019) held at Covenant University, Ota, Ogun State Nigeria (June 19-21, 2019).
- Coastal Ocean Environmental Summer School in Ghana (COESSING 2018), University of Ghana, Legon, Ghana. (July 29-August 4, 2018).
- International Symposium on Business and Industrial Statistics, ISBIS-2018, Athens, Greece. (July 4-6, 2018).
- 14th Biennial Conference of the Athenian Policy Forum, University of Piraeus, Athens, Greece. (July 6-8, 2018).
- A 3-Day Project-Based Learning Workshop organized by Center for Project Based Learning, Worcester Polytechnic Institute (WPI), USA 21-23 May 2018.
- CSAE 2018 Conference "Economic Development in Africa", Oxford University, Oxford, United Kingdom. (March 18-21, 2018).
- Project-Based Learning Workshop organized by Worcester Polytechnic Institute, MA, USA at Anchor University Lagos, Nigeria. (May 21-23, 2018).
- Practical Statistical Techniques for Collecting and Analyzing Data held in February, 27th -1st March, 2018 at Anchor University, Lagos.
- 1st National Big Data Economic Summit, Lekki Lagos, Nigeria (12th October 2017).
- 1st International Statistical Conference on Positioning National Statistical Systems for Data Revolution and Inclusive Development. (Organized by the Nigeria Statistical Association at the University of Lagos)- 6th-8th Sept. 2017. Lagos, Nigeria.
- 61st ISI World Statistics Congress held at Palais Des Congress, Marrakech, Morocco (16-21 July 2017).
- Applied Meta-Analysis using R held at Palais Des Congress, Marrakech, Morocco (15th July 2017).
- School on Recent Advances in Analysis of Multivariate Ecological Data: Theory and Practice, held at The Abdus-Salam International Centre for Theoretical Physics (ICTP), Trieste, Italy. (24-28 October 2016).
- 1st National Conference on Environment and Health, Redeemer's University, Ede, Osun State, Nigeria (17th -18th May 2016).
- American Statistical Association Conference on Statistical Practice, Tampa, FL, USA, (Feb. 22-24, 2015).
- 60th World Statistics Congress (ISI–International Statistical Institute), Rio Centro, Brazil. (July 26-31, 2015).
- 13th Faculty Seminar by Professor Eric Vance (Director, Laboratory for Interdisciplinary Statistical Analysis (LISA), Virginia Tech, USA), 'Sustaining Statistical Collaboration Laboratory at OAU', Faculty of Science, Obafemi Awolowo University, Ile-Ife, Nigeria. (14th January 2015).

- 39th Annual Conference of the Nigerian Statistical Association, Osogbo, Nigeria. (9- 11 September 2015).
- Joint Statistical Meetings, Boston, MA, USA (July 31-August 6, 2014).
- 92nd Annual Conference of the Virginia Academy of Science (VAS) held at the Virginia Commonwealth University, Richmond, VA, USA. (15-17 May 2014).
- Annual International Faculty of Science Conference 2014, Quest for Renewed Energy and Sustainable Development, Obafemi Awolowo University, Ile-Ife, Nigeria. (8th - 10th , October 2014).
- ISBA-George Box Research Workshop on Frontiers of Statistics held at George Washington University, Washington DC, USA. (20-22 May 2014).
- African Coalition Brownbag Series held at Virginia Tech, Blacksburg, VA, USA. (April 2nd, 2014).
- 8th PAN African Congress of Mathematicians held at National Mathematical Centre, Abuja, Nigeria. (July 31-August 8, 2013).
- International Conference on Objective Bayes (O-Bayes13) held at Duke University, North Carolina, Durham, USA. (Dec. 15-19, 2013).
- 32nd Annual Conference of the Nigeria Mathematical Society (Ife 2013) held at Obafemi Awolowo University, Ile-Ife, Nigeria. (June 25-28, 2013).
- Research Workshop on 'Confronting Intractability in Statistical Inference' organized by the Department of Statistics, University of Bristol, Bristol, United Kingdom. (16- 19, April 2012).
- Annual International Faculty of Science Conference 2012, "Science: The Bedrock for Human Development", Faculty of Science, Obafemi Awolowo University, Ile-Ife, Nigeria (14th –16th of October, 2012).
- Research Workshop on 'Causal Inference in Longitudinal Studies' held at the School of Mathematics, University of Bristol, United Kingdom. (10-13, April 2012).
- 3rd Student's Conference on Operations Research (SCOR 2012) held at the University of Nottingham, United Kingdom (April 20-22, 2012).
- Conference in honor of the scientific contributions of Professor David Bartholomew, organized by the Department of Statistics, London School of Economics and Political Science, London, United Kingdom (12-13, December, 2011).
- 5th International Conference on Computational and Financial Econometrics, held at the Senate House, University of London, United Kingdom (16-19 December 2011).
- Capacity Building Workshop on Computer Packages for Lecturers in Tertiary Institutions, held at National Mathematical Centre (NMC), Abuja, Nigeria (16-21, October 2011).
- 35th Annual Conference of the Nigerian Statistical Association, held at Akure, Ondo State, Nigeria. (19-23, September 2011).
- Annual Faculty of Science Conference, Obafemi Awolowo University, Ile-Ife, Nigeria, (23-28 October 2010).

Keynote Speeches

i. Inaugural AIMS Network Talk, paper presented to the Six Centers of the African Institute for Mathematical Sciences in Ghana, Rwanda, Cameroon, Senegal, South Africa and Tanzania. Paper presented: Wavelets meet machine Learning: A Multiresolution Perspective of Predictive Modeling.

ii. 3rd International Conference on Statistical Theory, Methods and Applications. University of Nigeria, Nsukka. 24th April, 2025. Paper Presented: Integrating Wavelet Transforms into Explainable Machine Learning Decisions.

iii. 1st Anointed University Worldwide Leadership Conference: Developing Effective Leadership (September 4-5, 2023).

iv. 18th International Conference on Statistical Sciences, Imperial College of Business Studies, Lahore, Pakistan: Using Statistics for Sustainable Development Goals and Human Development (Feb 18-20, 2021).

v. ADA Global Academy Research Mentorship Series (April, 2022): Practical Skills for Conducting Research.

vi. ADA Global Academy Research Mentorship Series (April, 2023): Practical Skills for Conducting Research.

GRADUATE SUPERVISION ACTIVITIES

I have supervised the following theses at the graduate level at the African Institute for Mathematical Sciences (AIMS) and Obafemi Awolowo University.

i. Alfred Seesa- Long Memory Properties and Fractional Integration of Malaria Time Series Data. (MSc Mathematical Epidemiology, 2026 - AIMS Cameroon)

ii. Grace Ateladaga ASURU- Comparative Performance of MARS and Random Forest for Blood Pressure Prediction from Wavelet-Transformed PPG Signals and Clinical Data. (MSc Data Science, 2025 - AIMS Cameroon)

iii. Fred Okondzua- Wavelet-Based Anomaly Detection in Stock Market Data Using Advanced Unsupervised Learning Models. (MSc Data Science, 2025 - AIMS Cameroon)

iv. Peter Njoronge Nwangi- Explainable Machine Learning Models for Health Data Prediction with Improved Accuracy. (MSc Data Science, 2024 - AIMS Cameroon)

v. Brian April- Novel Applications of Bayesian Additive Regression Model in Diamond Mining: A Comparative Study of Tree-Based Methods. (MSc. Data Science, 2024 - AIMS Cameroon)

vi. Sinenkhosi Mamba - Explainability of Digital Wallets' Fraud Detection Algorithms: Comparative Analysis of SHAP and Anchors Methods. (MSc. Data Science, 2024 - AIMS Cameroon)

vii. Diallo Ramatoulaye - Comparative analysis of Matthews correlation coefficient (MCC), F1 score, and balanced accuracy in binary classification evaluation. (MSc. Data Science, 2024 AIMS Senegal)

viii. Edalo Codjo - Machine learning prediction of multiclass credit score classification. (MSc. Data Science, 2024 AIMS Senegal)

ix. Ismaila Elimane Ly- Comparative Analysis of Time Series Models versus Artificial Neural Network for Predicting Climate Data. (MSc. Data Science, 2024 AIMS Senegal).

x. Alvine Fandio- Comparative Analysis of Supervised Machine Learning Algorithms for Predicting Imbalanced Health Data. (MSc. Data Science, 2024 AIMS Senegal)

xi. Anthonia Oluchukwu Njoku- Improving the Accuracy of Financial Bankruptcy Prediction Using Ensemble Learning Techniques. (MSc Data Science, 2023 - AIMS Cameroon)

xii. Malibongwe Tsabedze- A Comparative Analysis of Sampling Methods for Imbalanced Data Classification in Machine Learning with Health Applications. (MSc Data Science, 2023 AIMS Cameroon)

xiii. Ndikum Sally Ngonsah Obasi- Application of Machine Learning in Geotechnical Engineering: A Comparative Study of Soil Classification Algorithms. (MSc. Data Science, 2023 AIMS Cameroon)

xiv. Adeola, Mary Oluwayemisi (SCP/13/14/8/1563)- A Correlation Study of Environmental Factors and Cancer Prevalence in Ondo State, Nigeria: Impact of Radioactivity in Soil and Air. (MSc. Physics and Engineering Physics- Specialization in Medical and Environmental Physics), Obafemi Awolowo University (OAU), Ile-Ile, Nigeria (Co-supervision).

International Professional Offices

i. Vice President-International Association for Statistical Education (IASE)

ii. Council Member, International Statistical Institute (ISI)

iii. Vice President- LISA 2020 Global Network, UCBoulder, USA.

iv. President- ADA Global Acedemy.

v. Vice-President for Membership- International Society for Business and Industrial Statistics (ISBIS)

vi. Associate Editor- Journal of Applied Sciences- Environmental Statistics and Data Science (JAS: ESDS) (Taylor and Francis)

Consultancies

i. **09/2020-09/2024 Machine Learning Engineer/Data Management Consultant, Human Sciences Research Council, Cape Town, South Africa**

- Engaged in supervised machine learning and classification projects using National survey datasets
- Coordinated several data collation and machine learning projects
- Reviewed codes and provided one-to-one mentorship.
- Served as Biostatistician on several projects.
- Utilized algorithmic, data mining, AI, machine learning, and statistical tools to identify trends in data.
- Trained researchers on machine learning and Biostatistics
- Built custom machine learning algorithms and traditional methods.
- Developed machine learning solutions-Algorithms include KNN, LDA, SVM, Decision Trees, random forest, and Naïve Bayes.
- Model development and validation.
- Acted as lead programmer for assigned projects
- Developed data-driven insights and reports for actionable decisions
- Published co-authored articles/book in high-impact journals
- Other activities include Data analysis, interpretation, reporting, missing value imputation, data wrangling, feature selection, data visualization, and EDA.

ii. **03/2020 - 09/2020 Contract Biostatistician, TPKR Scientifics, LLC, San Antonio, Texas, USA**

- Set timelines and milestones to achieve quality deliverables, accountable for the development and execution of SAPs and DMPs, programming, and other deliverables; work with management to ensure internal resourcing supports timelines
- Assisted in managing project budgets, and supporting financial reporting for studies, identifying out-of-scope work, and providing leadership in statistical analysis.
- Provided protocol review and input such as sample size calculations, multiplicity/alpha, and statistical methodology sections.
- Provided randomization materials including randomization schedules
- Developed product specifications for analysis datasets
- Led and/or supported the planning, management, and execution of statistical and statistical programming activities for various tasks including Biostatistics and clinical trials.
- Provided sound statistical input on study/research design to meet project needs and regulatory scientific requirements.
- Developed and updated Statistical Analysis Plans.

- Analyzed and interpreted results from clinical trials
- Assisted in the development of a proposal used to profile data for SDTM and
- Assisted in the development and review of Statistical Analysis Plan (SAP)

iii. **2016- Training Consultant, International Institute for Tropical Agriculture (IITA)**

- Designed comprehensive training programs and curricula tailored to the organization's data management requirements.
- Created training materials, including documentation, manuals, and interactive modules.
- Conducted training sessions for employees at various levels, ensuring effective knowledge transfer.
- Utilized diverse training methods, such as workshops and hands-on exercises, to cater to different learning needs of participants from various African countries including Ghana, Congo, Nigeria, Togo and the Benin Republic.
- Familiarized employees with relevant data management tools and technologies.
- Guided the proper use of databases, analytics tools, and data visualization platforms, including R software.
- Educated employees on industry best practices for data management and governance.
- Ensured that training programs included information on data protection regulations and compliance requirements.
- Stayed updated on emerging trends and technologies in data management to enhance training content.
- Tailored training programs to meet the specific needs of different departments within the organization.
- Recognized that various teams may have unique data management challenges and requirements.
- Effectively communicated complex data management concepts to individuals with varying levels of technical expertise.
- Facilitated discussions and promoted a shared understanding of data-related concepts across the organization.

Selected Short Courses and Workshops Taught

i. A One-Day Short Course on Regression Methods organized by the Department of Statistics, Virginia Tech, USA- October 2013.

ii. A One-Day Short Course on Basics of R organized by Department of Statistics, Virginia Tech, USA- February 2014.

iii. A One-Day Short Course on Survey Design and Analysis organized by Department of Statistics, Virginia Tech, USA- 24/03/2014.

iv. A 5-Day Intensive Workshop on Practical Data Analysis with R organized by the Department of Mathematics, Obafemi Awolowo University, Ile-Ife, Nigeria (11-15, April 2015).

v. A 3-Day Intensive Workshop on Practical Data Analysis with R for Ecologists organized by the Department of Mathematics, Obafemi Awolowo University, Ile-Ife, Nigeria (June 1-3, 2016).

vi. Workshop for Researchers on Data Management with R, IITA, Ibadan (July 2016)

vii. A 2-Day Intensive Workshop on Practical Steps in Collecting and Analyzing Research Data organized by the Department of Mathematics, Anchor University, Lagos, Nigeria (September 2017).

viii. A 3-Day Intensive Workshop on Practical Steps in Collecting and Analysing Research Data organized by the Department of Mathematical Sciences, Anchor University, Lagos, Nigeria (Feb. 27-March 1, 2018).

ix. Data Analysis, Visualisation, and Programming with Python Workshop held at Anchor University Lagos, Nigeria (12-15/11/2018).

x. ISBIS Regional Workshop on Basic and Advanced Time Series Analysis: Theory, Practice, and Programming. Anchor University Lagos (September 11-14, 2018).

xi. A 2-Day Intensive Workshop on Article Writing, Publication and Grantsmanship, Anchor University Lagos, Nigeria (March 28-29, 2019).

xii. Statistical Data Analysis with R- African Biosciences Limited (ABL), Ibadan, Oyo State, Nigeria (June 10-12, 2019).

xiii. Problem-Project Based Learning Workshop-Held at Anchor University Lagos (June 13, 2019)

xiv. ISBIS Regional Workshop on Time Series Analysis with Big Data Applications. Anchor University Lagos. (September 9-11, 2019).

xv. Training on R, Python, and Machine Learning. Olusegun Agagu University of Science and Technology, Ondo. (July 25-27, 2023).

xvi. Workshop on Artificial Intelligence for Malaria (and Infectious Disease Modeling) AIMM 2024, IMECC, UNICAMP, Brazil (August 28-30, 2024).

xvii. A One-Day Short Course on Explainable Machine Learning in organized by the Department of Applied Mathematics, University of Colorado, Boulder, Boulder, October 2024.

Other Reviews

i. Reviewer, 7th Lindau Nobel Laureate Meeting in Economic Sciences, Lindau, Germany.

ii. Associate Editor- Journal of Applied Statistics- Environmental Statistics and Data Science (Taylor and Francis)

iii. Reviewer for impact factor journals like Expert Systems with Applications, Economics Letters, Statistical Papers, BMC Public Health, Expert Systems with Applications, Scientific Reports, BMC Pediatrics, African Economic Review, EJASA, SERJ, Statistics in Transition, etc.

iv. External examiner for MSc. Biostatistics Module on Biostatistical Collaboration-University of Stellenbosch, South Africa (2023 - Present).
I served as an external reviewer of the following theses:

A. Emilienne Rachel Kenko: Simulation of Mixed Brachytherapy Treatment by Monte Carlo Method. (MSc. Data Science, 2023-AIMS Senegal).

B. Amady Sy: Extreme value analysis of rainfall in West Africa: Case of Senegal. (MSc. Data Science, 2023- AIMS Senegal).

C. Adamu Abubakar Saad: Extreme Value Analysis of COVID-19 Levels in Africa. (MSc Data Science, 2023- AIMS Senegal).

Research Grants & Funding

Research Grants

@lXlr@ **Year Project / Grant Source Amount**
2021–2023 Research Grant, IMECC, UNICAMP FAPESP, Brazil US$40,000
2021 Book Publication Grant USAID / LISA 2020 US$15,000
2020 Statistical Survey on Wealth Creation, Ekiti State USAID / LISA 2020 TEACH Fund US$11,000
2013 Research on Econometric Modelling of the Nigerian Economy TETFund, Nigeria US$20,000
2013 Research Visit, Virginia Tech Google Research Grant US$40,000

Selected Fellowships & Travel Awards

@lXlr@ **Year Award Source Amount**
2019 Regional Workshop on Time Series, Sub-Saharan Africa World Bank / ISBIS US$8,500
2019 Travel Grant, LISA 2020 Symposium, Malaysia USAID US$3,000
2018 Travel Grant, CSAE Conference, Oxford University USAID US$3,500
2017 Travel Award, 61st World Statistics Congress, Morocco African Dev. Bank US$2,500
2015 Travel Award, 60th World Statistics Congress, Brazil World Bank US$2,500
2020 Travel Award, Research & Policy Workshop, Nairobi AFIDEP / AAS US$2,000
2016 Travel Grant, ISBIS Meeting, Barcelona Naval Research US$2,000
2013 Travel Grant, O-Bayes, Duke University ISBA US$2,000
2013 Travel Grant, ICOTS9, Flagstaff World Bank US$2,500
2014 Best Paper Runner-Up, Virginia Academy of Science Virginia Academy US$50